Math Inquirers

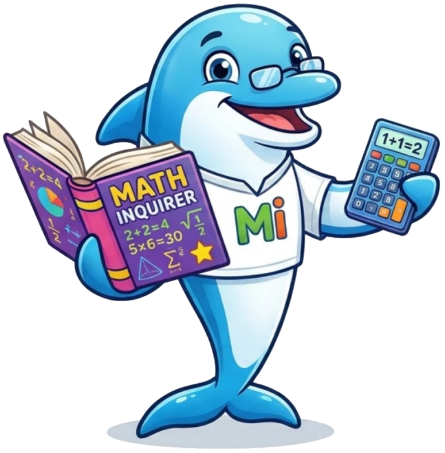
Think. Question. Prove.

Level 1 (ages 6-8) - Term 2

Workbook

Book 4

Lessons 16 to 20



My name:

Contents

Math Inquirers Level 1, Term 2 - Workbook, Book 4

Lesson 16 **Skip-Count Tables** page 3
Multiplication: Skip-Count Tables

Lesson 17 page 11

Lesson 18 **Times Stories** page 19
Multiplication And Division: Times Stories

Lesson 19 **Picture Equations** page 27
Reasoning About Equality: Picture Equations

Lesson 20 **Find One First** page 35
Division And Money: Find One First

Kit pages (cut-outs)

Some lessons end with a kit page: tens bars, number lines, cards and so on.
Each kit page has a blank back, so you can cut or tear it out without losing any work.
Tear along the dashed line in the left margin, then cut out the pieces. Keep them in an envelope.

SKIP-COUNT TABLES!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use your 100 square and a counter. In 3 × 5, the first number is how many jumps and the second is the size of each jump.

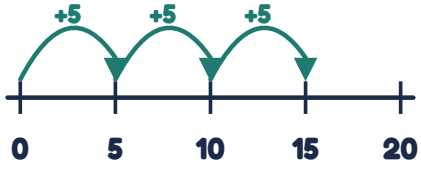
No.S1

Count in 5s. Fill in the missing numbers.

5	10	☐☐	20	☐☐	30	☐☐	40

No.S2

A frog jumps along the number line. Fill in the blanks.



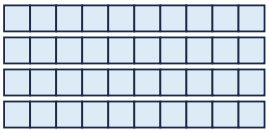
(a) The frog makes ☐ jumps.

(b) Each jump is ☐ long.

(c) ☐ × ☐ = ☐☐

No.S3

Ravi has 4 bundles of 10 straws. How many straws does Ravi have in all?



(a) 10 + 10 + 10 + 10 = ☐☐

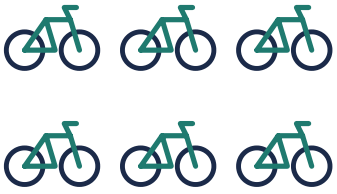
(b) 4 × 10 = ☐☐

Name _____ Date _____

Use your 100 square to count the jumps.

No.S4

Each bike has 2 wheels. How many wheels do 6 bikes have?



(a) 6 × 2 = ☐☐

(b) 6 bikes have ☐☐ wheels.

No.S5

Fill in the table.

Jumps of 5	Total
1 × 5	5
2 × 5	☐☐
4 × 5	☐☐
6 × 5	☐☐
8 × 5	☐☐

No.S6

Read each statement. Decide if it is True or False.

Statement	True	False
5 × 4 = 20	☐	☐
3 × 10 = 40	☐	☐
6 × 2 = 12	☐	☐
8 × 5 = 35	☐	☐

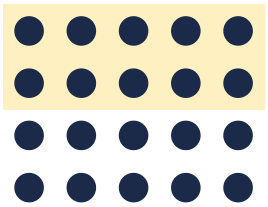
Put a tick (✓) in the correct box.

Name _____ Date _____

Use your 100 square or counters. Choose any question, in any order.

No.T1

Look at the dots. The top 2 rows are shaded. Fill in the blanks.



(a) 2 × 5 = ☐☐

(b) 4 × 5 = ☐☐

No.T2

Colour the 2s. Ring the 5s. Then fill in the blanks.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40

The numbers in both tables are ☐☐, ☐☐, ☐☐ and ☐☐.

No.T3

Is each number in the 5s table? Circle Yes or No.

(a) 50 **Yes** **No**

(b) 27 **Yes** **No**

(c) 65 **Yes** **No**

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Here are the 4s up to 40. Ring each number that is also in the 5s table. Then fill in the blanks.

4	8	12	16	20	24	28	32	36	40

Numbers in the 4s and the 5s: ☐☐ and ☐☐.

No.G2

Count in 3s up to 30. Fill in the missing numbers. Then find the numbers in both the 3s and the 5s.

3	6	9	☐☐	15	☐☐	21	☐☐	27	30

(a) The missing numbers are ☐☐, ☐☐ and ☐☐.

(b) In both the 3s and the 5s: ☐☐ and ☐☐.

No.G3

Ali jumps 5 each time. Mei jumps 4 each time. They both want to reach 40.



(a) Ali needs ☐ jumps.

(b) Mei needs ☐☐ jumps.

(c) Mei needs ☐ more jumps than Ali.

Name _____ Date _____

Investigation: Last digits

First, try this. Write the last digit of each number in the 5s up to 40: 5, 10, 15, 20, 25, 30, 35, 40.

Next, try this. Which of 10, 15, 20 and 25 are in both the 2s and the 5s? Write the last digit of each one you find. Write what you notice.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Count in 5s. Fill in the missing numbers.

5, 10, ☐☐, 20, ☐☐, 30

No.E2

Which sentence is true? Write its number in the bracket.

(1) 3 × 5 = 20

(2) 4 × 10 = 40

(3) 6 × 2 = 14

(4) 8 × 5 = 35

( )

No.E3

Ben puts 5 stickers on each of 6 pages. How many stickers does Ben have in all?

(a) ☐ × ☐ = ☐☐

(b) Ben has ☐☐ stickers in all.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ counting the jumps ☐ writing the facts ☐ finding the last digits ☐ something else

What helped me: ☐ the 100 square ☐ a counter as a frog ☐ drawing it ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name _____ Date _____

Kit page: 100 square

Nothing to cut. The square starts on 0: put your counter frog there. Colour lightly with pencil crayons.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

TEAR OUT here (blank on the back) - then cut out the pieces

SHARING AND GROUPING!

Name _____ Date _____

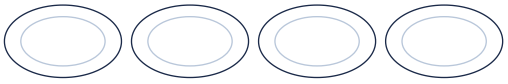


After question 3 you may choose: carry on, or try **Think harder**.

Use counters and plates, or draw it.

No.S1

Share 8 dots equally among the 4 plates. Draw the dots. Then fill in the blank.



(a) Each plate has ☐ dots.

No.S2

Mei packs 12 cookies into bags. She puts 3 cookies in each bag. Draw a ring around each group of 3. Then fill in the blank.



(a) Mei needs ☐ bags.

No.S3

Ali has 14 sweets. He shares them equally among 2 friends. How many sweets does each friend get?



(a) Each friend gets ☐ sweets.

Name _____ Date _____

Deal one at a time, or count in jumps.

No.S4

Share the sweets equally among the friends. Fill in the table.

Sweets	Friends	Each gets
8	2	☐
12	4	☐
20	5	☐

No.S5

Ben puts 20 pencils into pots. He puts 5 pencils in each pot. How many pots does Ben use?



(a) Ben uses ☐ pots.

No.S6

Mei shares 12 stickers equally among 4 friends. Each friend gets 3 stickers.

(a) Fill in the number sentence to check. ☐ × ☐ = ☐☐

(b) Does it match the 12 stickers? Circle Yes or No. **Yes** **No**

No.S7

Read each story. Circle Share or Group.

(a) Lily gives 12 beads equally to 3 friends. **Share** **Group**

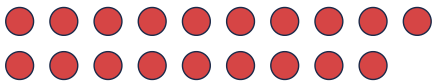
(b) Jamie puts 12 beads into boxes of 6. **Share** **Group**

Name _____ Date _____

Use counters or draw it. Choose any question, in any order.

No.T1

Ali has 19 counters. He puts them into groups of 4.



(a) Ali makes ☐ groups of 4.

(b) There are ☐ counters left over.

No.T2

Ben shares 20 biscuits equally with his 3 cousins. Aisha says 3 people share the biscuits.

(a) Is Aisha correct? Circle Yes or No. **Yes** **No**

(b) Each person gets ☐ biscuits.

No.T3

Mei has 18 sweets. Read the two stories.

(a) Story A: Mei puts the 18 sweets into bags of 6. Circle Share or Group. **Share** **Group**

(b) Mei needs ☐ bags.

(c) Story B: Mei shares the 18 sweets equally among 6 bags. Circle Share or Group. **Share** **Group**

(d) Each bag has ☐ sweets.

No.T4

Ravi has 17 stickers. He puts 5 stickers in each bag.

(a) He fills ☐ bags.

(b) There are ☐ stickers left over.

(c) He needs ☐ more stickers to fill one more bag.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Jenny has 12 sweets. She shares them equally among some friends. Fill in the table. Then answer (a) and (b).

Friends	1	2	3	4	6	12
Each gets	☐☐	☐	☐	☐	☐	☐

(a) Can the 12 sweets be shared equally among 5 friends, with none left over? Circle Yes or No. **Yes** **No**

(b) If Jenny shares them among 5 friends, there are ☐ sweets left over.

No.G2

Jenny shares 20 pencils equally among 4 pots. Then she puts 3 more pencils in each pot.

(a) Each pot has ☐ pencils at first.

(b) Each pot has ☐ pencils now.

(c) There are ☐☐ pencils in all now.

No.G3

Wei has 20 stickers. He keeps 2 stickers. He shares the rest equally among 3 friends.

(a) Wei shares ☐☐ stickers.

(b) Each friend gets ☐ stickers.

Name _____ Date _____

Investigation: Sharing among 4 plates

First, try this. Put 8 counters on 4 plates, one at a time. How many on each plate? Now try 9 counters. How many are left over?

Next, try this. Share 13, 14, 15, 16, 17, 18, 19 and 20 counters equally among 4 plates. Write how many are left over each time. What do you notice?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

20 sweets are shared equally among 4 children. Fill in the blank.

(a) Each child gets ☐ sweets.

No.E2

Which story puts 12 sweets into groups of 4? Write its number in the bracket.

(1) Mei shares 12 sweets equally among 4 friends.

(2) Mei puts 12 sweets into bags of 4.

(3) Mei has 12 sweets and eats 4.

(4) Mei has 12 sweets and gets 4 more.

( )

No.E3

Ali puts 18 cards into piles. Each pile has 3 cards.

(a) Ali makes ☐ piles.

(b) Check: ☐ × 3 = 18

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ share or group ☐ dealing out ☐ how many groups ☐ something else

What helped me: ☐ counters on plates ☐ drawing it ☐ counting in jumps ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking it ☐ asking a question

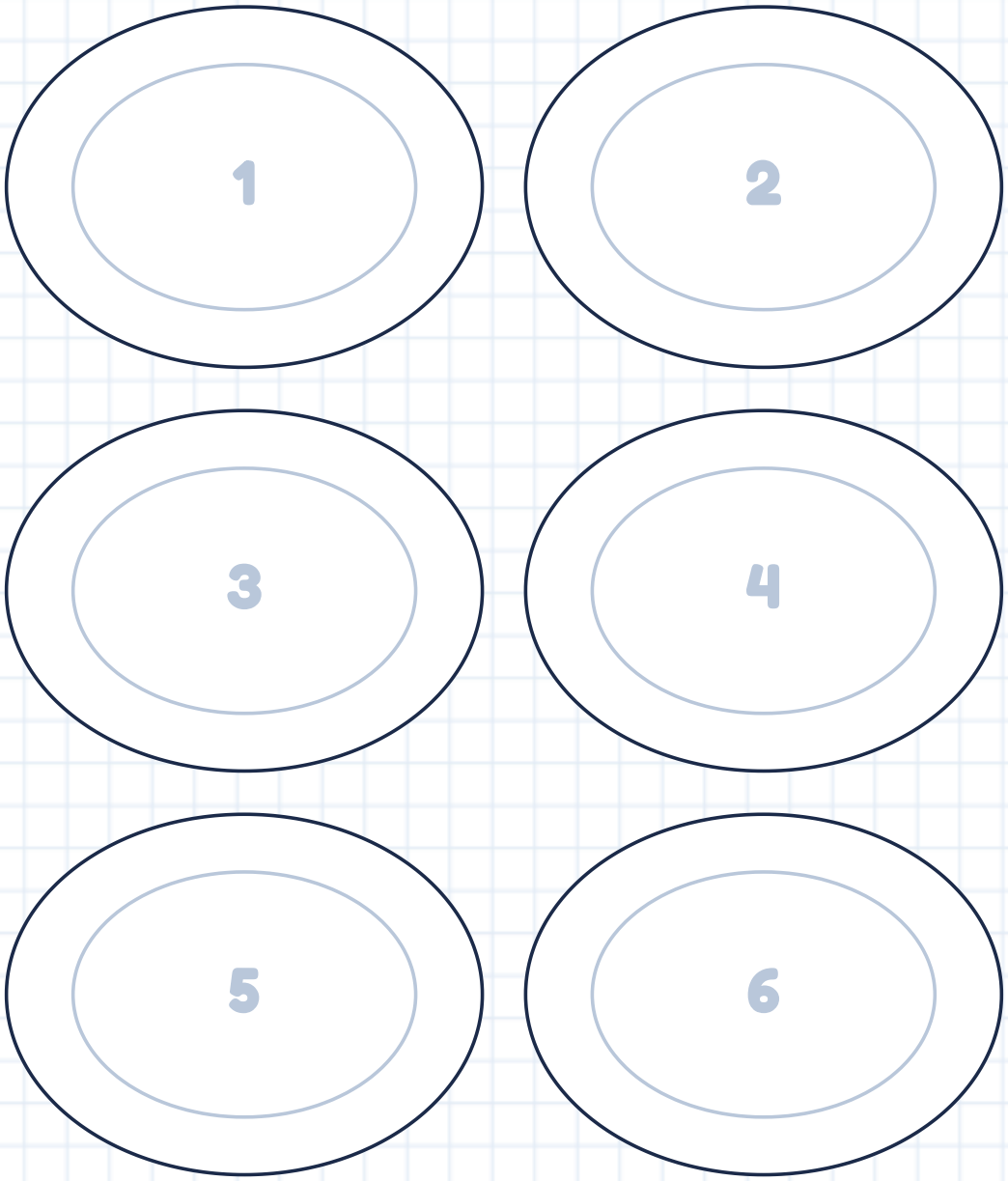
☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name _____ Date _____

Kit page: groups mat

Nothing to cut. Lay counters on the plates. Use it if you do not have plates or cups.



TEAR OUT here (blank on the back) - then cut out the pieces

TIMES STORIES!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

Use cups and counters, or draw the strip. Every box is the same size.

No.S1

Mei has 3 bags. Each bag has 5 sweets. Write the number in each box of the strip. Then fill in the blank.



(a) Mei has ☐☐ sweets in all.

No.S2

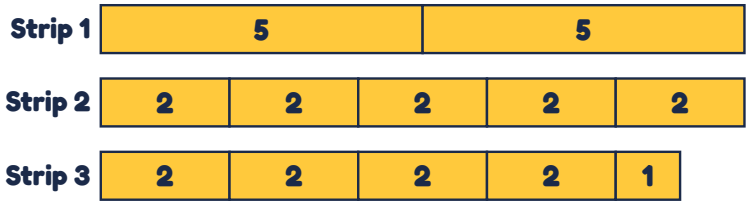
Ben has 4 boxes. Each box has 3 pencils. How many pencils does Ben have in all?

(a) ☐ groups of ☐ = ☐☐

(b) Ben has ☐☐ pencils in all.

No.S3

Siti has 5 bags. Each bag has 2 sweets. Which strip shows Siti's sweets? Write its number in the bracket.



(1) Strip 1

(2) Strip 2

(3) Strip 3

( )

Name _____ Date _____

Draw the strip if it helps. Do one step at a time.

No.S4

Ali has 3 boxes with 5 pens in each box. He also has 2 loose pens. How many pens does Ali have in all?

Pens	5	5	5	2

(a) The boxes have ☐☐ pens.

(b) Ali has ☐☐ pens in all.

No.S5

Mei has 2 bags. Each bag has 6 sweets. Draw a strip. Then fill in the blank.

(a) Mei has ☐☐ sweets in all.

No.S6

Ravi has 4 bags with 3 marbles in each bag. He loses 2 marbles. How many marbles does Ravi have now?

Marbles	3	3	3	3

(a) The bags have ☐☐ marbles.

(b) Ravi has ☐☐ marbles now.

Name _____ Date _____

Use strips or counters. Choose any question, in any order.

No.T1

Jenny has 3 boxes with 6 pens in each box and 5 loose pens. She gives 8 pens away.

(a) The boxes have ☐☐ pens.

(b) Jenny has ☐☐ pens in all.

(c) Jenny has ☐☐ pens left.

No.T2

Mei has 7 marbles. Ali has twice as many marbles as Mei.



(a) Ali has ☐☐ marbles.

(b) They have ☐☐ marbles together.

No.T3

Mei has 4 stickers. Ravi has twice as many stickers as Mei. Jenny says Ravi has 6 stickers.

(a) Is Jenny correct? Circle Yes or No. **Yes** **No**

(b) Ravi has ☐☐ stickers.

No.T4

Ali has twice as many cards as Mei. Ali has 24 cards. Ali gives some cards to Mei. Then they have the same number of cards.



(a) Mei has ☐☐ cards at first.

(b) Ali gives ☐ cards to Mei.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Wei puts 3 stickers on each of 4 cards. He puts 2 stickers on each of 5 other cards. He also has 4 stickers in his hand.

(a) The first 4 cards have ☐☐ stickers.

(b) The other 5 cards have ☐☐ stickers.

(c) Wei had ☐☐ stickers at first.

No.G2

Siti has 5 marbles. Ravi has 3 times as many marbles as Siti.



(a) Ravi has ☐☐ marbles.

(b) They have ☐☐ marbles altogether.

No.G3

Kumar has 3 times as many pens as Lily. They have 24 pens together.

(a) Lily has ☐ pens.

(b) Kumar has ☐☐ pens.

Name _____ Date _____

Investigation: Twice, and twice again

First, try this. Mei has 3 stickers. Ali has twice as many. How many stickers does Ali have? Now Ben has twice as many as Ali.

Next, try this. Jenny has twice as many stickers as Ben. Write how many stickers Mei, Ali, Ben and Jenny each have. What do you notice?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Ali has 4 bags. Each bag has 5 sweets. Which number sentence finds how many sweets Ali has in all? Write its number in the bracket.

(1) 4 + 5

(2) 4 × 5

(3) 5 − 4

(4) 4 + 4 + 4

( )

No.E2

Lily has 2 bags with 5 marbles in each bag. She also has 3 loose marbles. How many marbles does Lily have in all?

(a) Lily has ☐☐ marbles in all.

No.E3

Ravi has 3 bags with 4 sweets in each bag. He eats 2 sweets. How many sweets does Ravi have now?

(a) Ravi has ☐☐ sweets now.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: _____

The hardest part was ... ☐ drawing the strip ☐ the second step ☐ twice as many ☐ something else

What helped me: ☐ cups and counters ☐ drawing it ☐ one step at a time ☐ I was not stuck

Next time I will try: ☐ drawing it ☐ using objects ☐ checking it ☐ asking a question

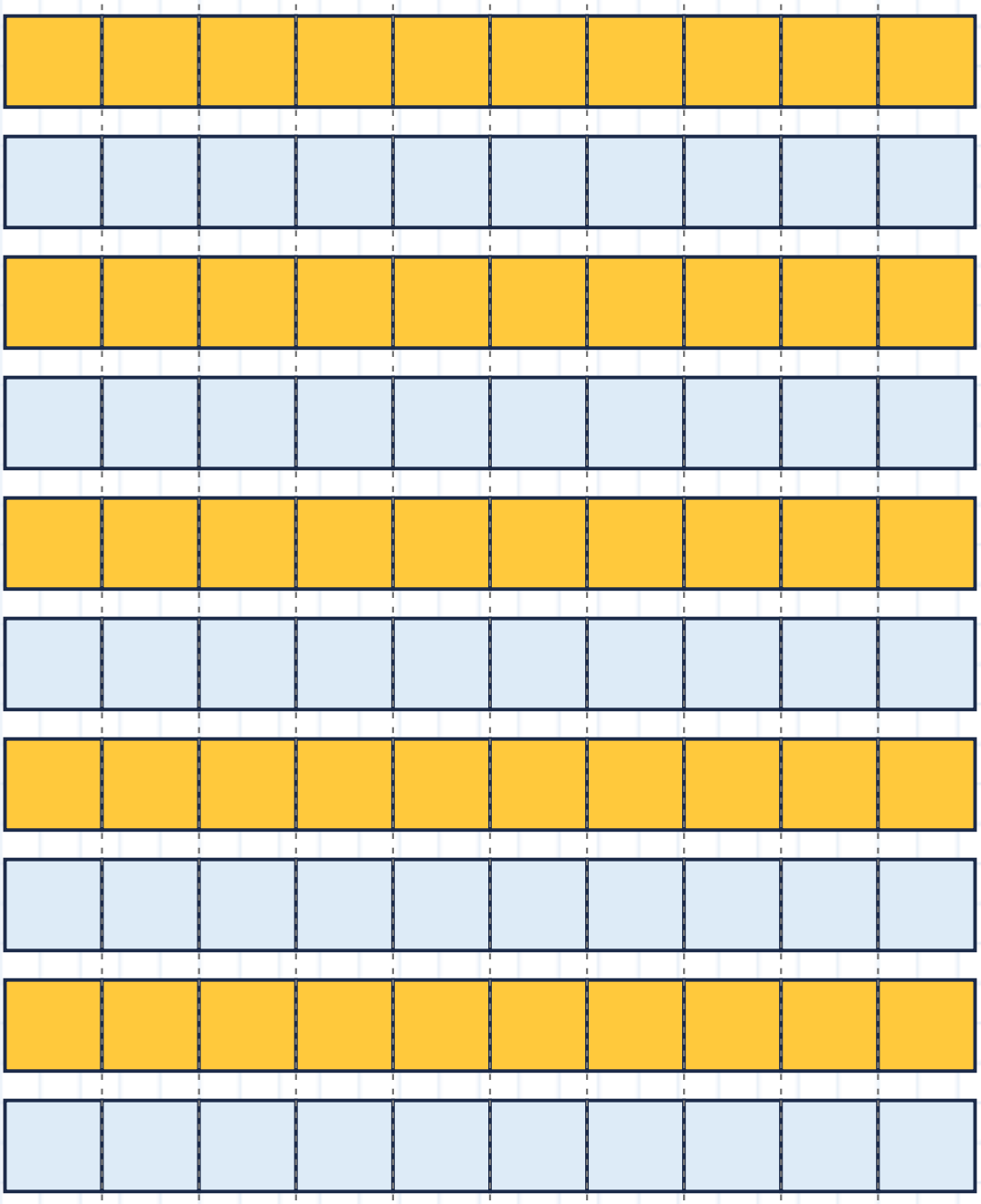
☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name _____ Date _____

Kit page: strips

A grown-up cuts out the ten strips before the lesson (cut between the strips, not between the cells) and cuts three of them after 6 cells.



TEAR OUT here (blank on the back) - then cut out the pieces

PICTURE EQUATIONS!

Name _____ Date _____

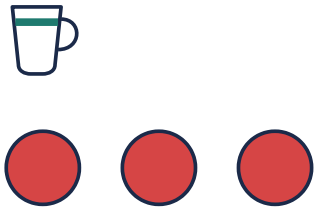


After question 4 you may choose: carry on, or try **Think harder**.

Each shape is a number. Use counters. Check your answer in the line.

No.S1

A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.



The cup holds ☐ counters.

No.S2

Each shape is a number. Find the number for the circle.

● + ■ = 10

■ = 4 ● = ☐

No.S3

Each shape is a number. Find the number for the heart.

♥ + ♥ = 14

♥ = ☐

No.S4

Each shape is a number. Find the number for the star.

★ + ★ + ★ = 15

★ = ☐

Name _____ Date _____

The same shape is always the same number. Different shapes are different numbers.

No.S5

Each shape is a number. The square is 4. Fill in the blanks.

△ + △ + ■ = 14

(a) The 2 triangles make ☐☐.

(b) Each triangle is ☐.

No.S6

Ben says the circle is 7. Look at the picture.

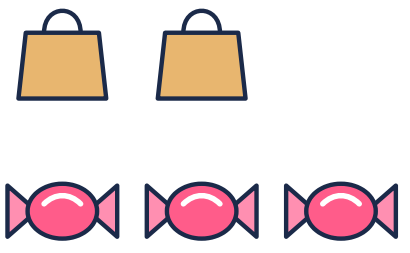
● + ● = 12

(a) Is Ben correct? Circle Yes or No. **Yes** **No**

(b) The circle is ☐.

No.S7

Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.



(a) The 2 bags have ☐☐ sweets in all.

(b) Each bag has ☐ sweets.

REASONING ABOUT EQUALITY: PICTURE EQUATIONS

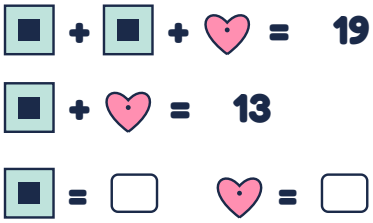
Think harder

Name _____ Date _____

Use counters or draw. Check your answer in every line. Choose any question.

No.T1

Each shape is a number. Find the number for each shape. Show your working.



No.T2

Kumar says the heart is 8 and the star is 4. Look at the picture. Decide if each statement is True or False.

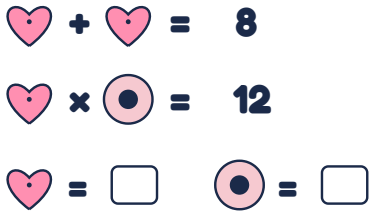


Statement	True	False
Line 1 is true with Kumar's numbers.	☐	☐
Line 2 is true with Kumar's numbers.	☐	☐

Put a tick (✓) in the correct box.

No.T3

Each shape is a number. Find the number for each shape.

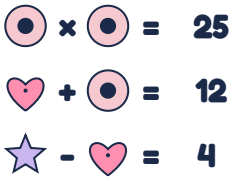


Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

Each shape is a number. Find the number for each shape.



(a) The circle is ☐.

(b) The heart is ☐.

(c) The star is ☐.

No.G2

Each shape is a number. Fill in the blanks.



(a) One star and one heart make ☐☐.

(b) The star is ☐.

(c) The heart is ☐.

No.G3

Each shape is a number. The heart is greater than the triangle. Find the number for each shape.



(a) The heart is ☐.

(b) The triangle is ☐.

REASONING ABOUT EQUALITY: PICTURE EQUATIONS

Choose, then look back

Name _____ Date _____

Investigation: One line or two?

First: The circle + the triangle = 10. One pair: circle 2, triangle 8. Write 2 more.

Next: Also: the circle + the circle + the triangle = 14. Which pairs fit both lines? Why?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Each shape is a number. The triangle + the triangle = 16. Find the number for the triangle.

The triangle is ☐.

No.E2

The heart + the square = 12. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.

(1) 5

(2) 6

(3) 7

(4) 8

( )

No.E3

Each shape is a number. The star is 5. Find the number for each circle.



Each circle is ☐.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well:

The hardest part was ... ☐ finding the number for one shape ☐ using two lines ☐ checking in every line ☐ something else

What helped me: ☐ counters and shape cards ☐ drawing it ☐ checking my answer ☐ I was not stuck

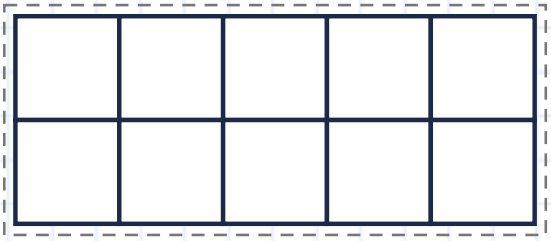
Next time I will try: ☐ using counters ☐ checking my answer ☐ drawing it ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Name _____ Date _____

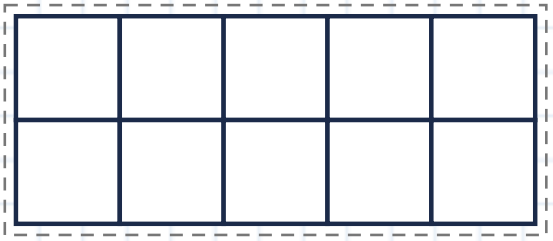
Kit page: ten-frames and counters

A grown-up cuts out the 20 counters (10 red, 10 yellow) before the lesson. Red counters can hide in a cup; yellow counters can be the loose ones. The ten-frames stay on the page.

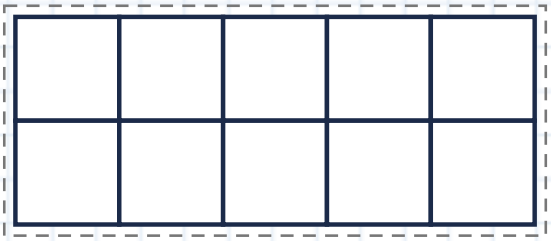
Ten-frame 1



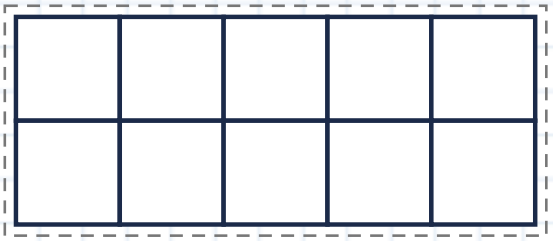
Ten-frame 2



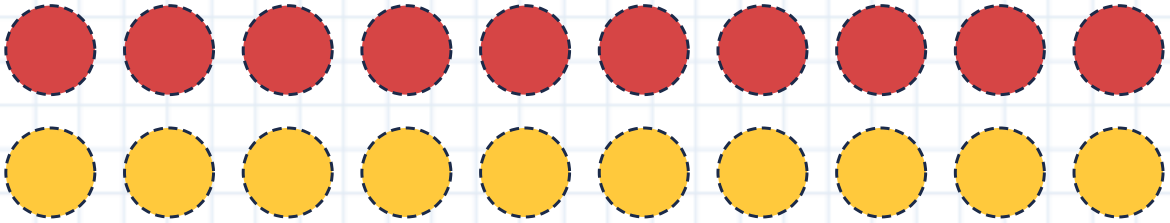
Ten-frame 3



Ten-frame 4



Counters: cut out all 20 (10 red, 10 yellow)



TEAR OUT here (blank on the back) - then cut out the pieces

FIND ONE FIRST!

Name _____ Date _____



After question 3 you may choose: carry on, or try **Think harder**.

A token is $1. Share the tokens to find 1. Count groups to find many.

No.S1

4 pens cost $20. How much does 1 pen cost?



1 pen costs $ ☐ .

No.S2

1 pen costs $3. How much do 4 pens cost? Fill in the blanks.



(a) 4 × 3 = ☐☐

(b) 4 pens cost $☐☐.

No.S3

6 buns cost $12. How much does 1 bun cost? Look at the strip.



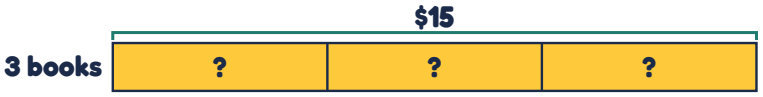
1 bun costs $ ☐ .

Name _____ Date _____

Find 1 first. Then find many. Each means for one.

No.S4

3 books cost $15. Fill in the blanks. Look at the strip.



(a) 1 book costs $ ☐ .

(b) 2 books cost $ ☐☐ .

No.S5

4 pens cost $12. Lily says 1 pen costs $4. Look at the picture.



(a) Is Lily correct? Circle Yes or No. **Yes** **No**

(b) 1 pen costs $ ☐ .

No.S6

Each book costs $3. Kumar buys 5 books. How much do the 5 books cost?



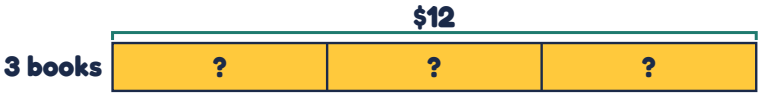
The 5 books cost $ ☐☐ .

Name _____ Date _____

Use tokens or draw a strip. Find 1 first. Choose any question.

No.T1

3 books cost $12. Fill in the blanks. Look at the strip.



(a) 1 book costs $ ☐ .

(b) 7 books cost $ ☐☐ .

No.T2

A dozen buns cost $24. Fill in the blanks.

(a) 1 bun costs $ ☐ .

(b) 3 buns cost $ ☐ .

No.T3

Wei pays $18 for 6 such cups. Fill in the blanks.

(a) 1 cup costs $ ☐ .

(b) 4 such cups cost $ ☐☐ .

No.T4

5 pens cost $15. Ben has $12. Fill in the blanks.

(a) 1 pen costs $ ☐ .

(b) Ben can buy ☐ pens.

Name _____ Date _____

Choose any question. You do not have to finish them all.

No.G1

2 hats cost $10. 3 scarves cost $18. Jenny buys 1 hat and 2 scarves. Fill in the blanks.

(a) 1 hat costs $ ☐ .

(b) 1 scarf costs $ ☐ .

(c) Jenny pays $ ☐☐ .

No.G2

6 pens cost $18. Mei has $20. She buys 4 pens. Fill in the blanks.

(a) 1 pen costs $ ☐ .

(b) 4 pens cost $ ☐☐ .

(c) Mei has $ ☐ left.

No.G3

Shop A sells 4 pens for $12. Shop B sells 5 pens for $20. Which shop sells 1 pen for less?

Shop	Pens	Cost in $
A		12
B		20

(a) At Shop A, 1 pen costs $ ☐ .

(b) At Shop B, 1 pen costs $ ☐ .

(c) Circle Shop A or Shop B. **Shop A** **Shop B**

Name _____ Date _____

Investigation: Which pack costs less?

First: A pack of 2 pens costs $6. How much does 1 pen cost?

Next: A pack of 5 pens costs $15. Does 1 pen cost less in this pack? Why?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

3 pencils cost $9. How much does 1 pencil cost?

1 pencil costs $ ☐ .

No.E2

Each sticker costs $2. How much do 5 stickers cost? Write its number in the bracket.

(1) $7

(2) $10

(3) $3

(4) $25

( )

No.E3

4 buns cost $12. Fill in the blanks.

(a) 1 bun costs $ ☐ .

(b) 3 buns cost $ ☐ .

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well:

The hardest part was ... ☐ sharing to find one ☐ finding many from one ☐ using the strip ☐ something else

What helped me: ☐ tokens ☐ drawing a strip ☐ checking my answer ☐ I was not stuck

Next time I will try: ☐ using tokens ☐ drawing a strip ☐ checking my answer ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

Name _____ Date _____

Kit page: $1 tokens

A grown-up cuts out the 20 round tokens before the lesson. A token stands for one dollar. Whole dollars only: there are no coins or notes in this lesson.



TEAR OUT here (blank on the back) - then cut out the pieces